

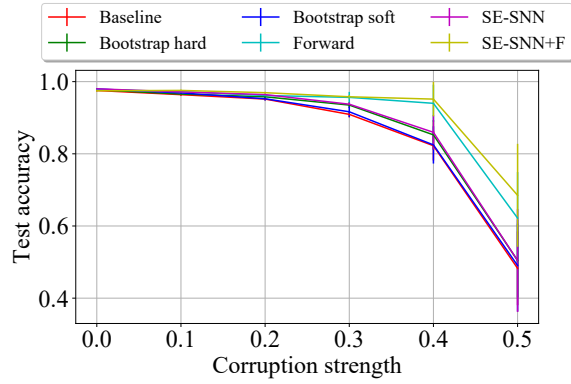


Figure 3: Label noise results on MNIST

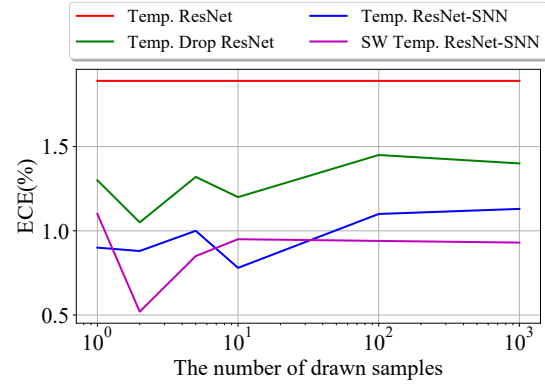


Figure 4: Calibration Results with ResNet on CIFAR-100

Dataset	CIFAR-10				CIFAR-100			
Model	ResNet	Res-SNN	WRN	WRN-SNN	ResNet	Res-SNN	WRN	WRN-SNN
Temp.	1.39	1.07	0.90	0.79	1.89	1.43	2.49	1.86
Temp. Drop	1.09	-	0.72	-	1.13	-	2.03	-
SW Temp.	-	0.90	-	0.39	-	0.93	-	1.64

Table 4: ECE (%) with 15 bins on ResNet-110 (ResNet), Wide-ResNet-16-2(WRN), and their corresponding SE-SNN version (e.g., Res-SNN for ResNet) with temperature scaling (Temp.), temperature scaling of dropout (Temp. Drop) in (Gal and Ghahramani 2016), and sample-wise temperature scaling (SW Temp.) using the per-sample variance learned by SE-SNN.

SE-SNN is already very competitive – it is only beaten by Forward Correction when the corruption strength (percentage of samples with label noise) is large (> 0.2). Once our SE-SNN is combined with the same procedure adopted by Forward Correction (SE-SNN+Forward), we achieve the best result.

Calibration

Experimental Setting Following the architecture and dataset used in (Guo et al. 2017), we train ResNet-110 and Wide-ResNet-16-2 on CIFAR-10 and CIFAR-100. For SE-SNN, a stochastic layer is added before the classification layer in these two neural networks respectively. Expected Calibration Error (ECE) (Naeini, Cooper, and Hauskrecht 2015) is the widely-used evaluation matrix to measure the calibration error. ECE approximates the difference in expectation between confidence and accuracy by partitioning predictions into K equally-spaced bins and taking a weighted average of the bins’ difference between accuracy and confidence. Specifically, ECE computes:

$$ECE = \sum_{k=1}^K \frac{|B_k|}{n} |acc(B_k) - conf(B_k)|, \quad (10)$$

where n is the number of samples, B_k denotes the set of samples in bin k . acc and $conf$ respectively denote the average accuracy and confidence (e.g., Eq. 8) of items in the specified bin.

Results Using SE-SNN we can draw multiple samples from the distribution corresponding to a single input, and

use these to produce an average softmax result for prediction. The ability to average over multiple samples should lead to an improved calibration (as per other approaches to Bayesian neural networks (Gal and Ghahramani 2016), which is denoted as ‘Temp. Drop’.), even before using our novel sample-wise temperature scaling strategy in Eq 8. Fig. 4 shows the ECE result for ResNet-110 on CIFAR100 as a function of the number of samples drawn. It can be seen that, as a function of the number of samples drawn to estimate probability posterior p_i , performance stabilizes after 10 samples. Table 4 summarizes the results for CIFAR-10/100 using ResNet-110 and WideResNet WRN-16-2 and 100 samples. We can see that: (1) Temperature scaled ResNet-SNN improves performance compared to temperature scaled vanilla ResNet. (2) Our sample-wise temperature scaling (‘SW Temp’) further improves performance compared to vanilla temperature scaling, reducing the calibration error by 12-50% relatively to vanilla temperature scaling. (3) Our method is clearly more effective than the dropout temperature scaling based one in (Gal and Ghahramani 2016) (‘SW Temp’ vs. ‘Temp. Drop’).

Conclusion

We proposed a simple and effective stochastic neural network framework. Our model is related to VIB and variational dropout, but provides a simpler and more direct realization via neuron regularization by a non-informative activation prior. Our extensive experiments show that this simple framework has diverse benefits for network pruning, adversarial defense, label noise robust learning, and model calibration.

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